# Chapter 7. Stokes' Theorem

Integrating forms over manifolds, orientations, the generalized Stokes' theorem and its applications to vector analysis

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## Overview

Chapter 6 showed how $k$-forms generalize scalar and vector fields in $\mathbb{R}^3$, and how $d$ generalizes grad, curl and div. We now define the integral of a $k$-form over a $k$-manifold, generalizing line and surface integrals in $\mathbb{R}^3$; these appear in the classical Stokes' theorem and divergence theorem, and integrals of forms appear in the generalized Stokes' theorem (§37). Recall the convention: all manifolds, forms, vector fields and scalar fields are of class $C^\infty$.

## §33. Integrating Forms over Parametrized-Manifolds

**Motivation.** A $k$-form can be integrated over a parametrized $k$-manifold by pulling it back to $\mathbb{R}^k$. The change of variables theorem shows that the result depends only on the orientation of the parametrization.

**Definition 33.1 (integral of a $k$-form over an open set of $\mathbb{R}^k$)**

Let $A$ be open in $\mathbb{R}^k$ and $\eta$ a $k$-form defined in $A$; $\eta$ can be written uniquely as $\eta = f\,dx_1 \wedge \cdots \wedge dx_k$. The **integral of $\eta$ over $A$** is

$$\int_A \eta = \int_A f,$$

provided the latter integral exists.

This seems coordinate-dependent, since $dx_i$ depends on the standard basis of $\mathbb{R}^k$. But if $\mathbf{a}_1, \ldots, \mathbf{a}_k$ is *any* right-handed orthonormal basis of $\mathbb{R}^k$, one checks $\int_A \eta = \int_{\mathbf{x}\in A} \eta(\mathbf{x})((\mathbf{x};\mathbf{a}_1), \ldots, (\mathbf{x};\mathbf{a}_k))$; so the integral depends on the orientation of $\mathbb{R}^k$ but not on the choice of basis.

**Definition 33.2 (integral over a parametrized-manifold)**

Let $A$ be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^n$ of class $C^\infty$, and $Y = \alpha(A)$, so that $Y_\alpha$ is a parametrized-manifold. If $\omega$ is a $k$-form defined in an open set of $\mathbb{R}^n$ containing $Y$, the **integral of $\omega$ over $Y_\alpha$** is

$$\int_{Y_\alpha} \omega = \int_A \alpha^*\omega,$$

provided the latter integral exists. It is linear in $\omega$.

The integral is invariant under reparametrization *up to sign*:

**Theorem 33.1**

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^k$. Assume $\det Dg$ does not change sign on $A$. Let $\beta : B \to \mathbb{R}^n$ be a map of class $C^\infty$; let $Y = \beta(B)$. Let $\alpha = \beta \circ g$; then $\alpha : A \to \mathbb{R}^n$ and $Y = \alpha(A)$. If $\omega$ is a $k$-form defined in an open set of $\mathbb{R}^n$ containing $Y$, then $\omega$ is integrable over $Y_\beta$ if and only if it is integrable over $Y_\alpha$; in this case,

$$\int_{Y_\alpha} \omega = \pm \int_{Y_\beta} \omega,$$

where the sign agrees with the sign of $\det Dg$.

**Proof.** Let $\mathbf{x}$, $\mathbf{y}$ denote the general points of $A$, $B$. We must show $\int_A \alpha^*\omega = \epsilon \int_B \beta^*\omega$, with $\epsilon = \pm 1$ the sign of $\det Dg$. Put $\eta = \beta^*\omega$; since $\alpha^*\omega = g^*(\beta^*\omega)$, we need $\int_A g^*\eta = \epsilon \int_B \eta$. Write

$\eta = f\,dy_1 \wedge \cdots \wedge dy_k$. By Theorem 32.2 (with $k = n$),

$$g^*\eta = (f \circ g)g^*(dy_1 \wedge \cdots \wedge dy_k) = (f \circ g)\det(Dg)\,dx_1 \wedge \cdots \wedge dx_k,$$

so we need $\int_A (f \circ g)\det Dg = \epsilon \int_B f$. This is the change of variables theorem, since $\det Dg = \epsilon|\det Dg|$. ■

If $A$ is connected, the hypothesis that $\det Dg$ does not change sign is automatic: $\{\det Dg > 0\}$ and $\{\det Dg < 0\}$ are disjoint open sets with union $A$.

**Theorem 33.2**

Let $A$ be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class $C^\infty$; let $Y = \alpha(A)$. Let $\mathbf{x}$ denote the general point of $A$; and let $\mathbf{z}$ denote the general point of $\mathbb{R}^n$. If $\omega = f\,dz_I$ is a $k$-form defined in an open set of $\mathbb{R}^n$ containing $Y$, then

$$\int_{Y_\alpha} \omega = \int_A (f \circ \alpha)\det(\partial\alpha_I/\partial\mathbf{x}).$$

**Proof.** By Theorem 32.2, $\alpha^*\omega = (f \circ \alpha)\det(\partial\alpha_I/\partial\mathbf{x})\,dx_1 \wedge \cdots \wedge dx_k$. ■

The notion of a $k$-form, and even more of its integral over a parametrized-manifold, is abstract; §36 gives a geometric interpretation.

**The "$dx$" notation of calculus**

If $\eta = f\,dx$ is a 1-form on $A = (a, b)$, then $\int_A \eta = \int_A f$, i.e. $\int_A f\,dx = \int_a^b f$: the left side is the integral of a *form*, the right of a *function*, and they are equal by definition. Line integrals also make sense: for a 1-form $P\,dx + Q\,dy + R\,dz$ on an open set of $\mathbb{R}^3$ and a parametrized-curve $\gamma : (a, b) \to \mathbb{R}^3$ with image $C$, Theorem 33.2 gives

$$\int_{C_\gamma} P\,dx + Q\,dy + R\,dz = \int_{(a,b)} \Big[P(\gamma(t))\frac{d\gamma_1}{dt} + Q(\gamma(t))\frac{d\gamma_2}{dt} + R(\gamma(t))\frac{d\gamma_3}{dt}\Big],$$

the calculus formula. The "$dx\,dy$" notation for double integrals is harder: $\iint_A f(x, y)\,dx\,dy$ resembles both the iterated integral, $\int_c^d[\int_a^b f(x, y)\,dx]\,dy = \int_A f$, and the integral of a 2-form, $\int_A f\,dx \wedge dy = \int_A f$. Reversing $x$ and $y$, the iterated integral still gives $\int_A f$ but $\int_A f\,dy \wedge dx = -\int_A f$. To avoid confusion we do not use "$dx\,dy$". The "$dV$" notation of Chapter 5 is unambiguous: if $A$ is open in $\mathbb{R}^k$, parametrized by the identity, then $\int_{A_\alpha} f\,dV = \int_A (f \circ \alpha)V(D\alpha) = \int_A f$; the $d$ there has nothing to do with the operator $d$.

## §34. Orientable Manifolds

**Motivation.** To integrate forms over a whole manifold we need all patches to have compatible orientations. Orientability is the condition that this is possible; the Möbius band is the standard example where it fails.

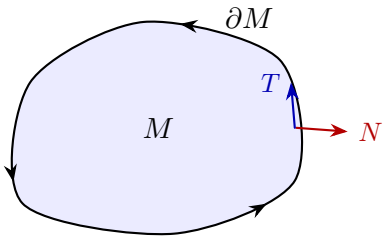



**Figure.** *The induced orientation on $\partial M$ for a region $M$ in $\mathbb{R}^2$: with $N$ the outward normal, the tangent $T$ is chosen so that $(N, T)$ is positively oriented, i.e. the boundary is traversed with $M$ on the left. This is the convention under which Green's and Stokes' theorems hold without sign changes.*

For $\omega$ supported in a single patch $\alpha : U \to V$ on $M$ we want $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega$. Since this is invariant under reparametrization only up to sign, we need an extra condition on $M$: *orientability*.

**Definition 34.1 (orientation-preserving diffeomorphism)**

A diffeomorphism $g : A \to B$ of open sets in $\mathbb{R}^k$ is **orientation-preserving** if $\det Dg > 0$ on $A$, and **orientation-reversing** if $\det Dg < 0$ on $A$.

This generalizes §20: $g$ is orientation-preserving iff for each $\mathbf{x}$ the linear transformation of $\mathbb{R}^k$ with matrix $Dg(\mathbf{x})$, i.e. $g_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{g(\mathbf{x})}(\mathbb{R}^k)$, is orientation-preserving.

**Definition 34.2 (orientable manifold, orientation)**

Let $M$ be a $k$-manifold in $\mathbb{R}^n$. Coordinate patches $\alpha_i : U_i \to V_i$ $(i = 0, 1)$ on $M$ **overlap positively** if the transition function $\alpha_1^{-1} \circ \alpha_0$ is orientation-preserving (when $V_0 \cap V_1 \neq \emptyset$). If $M$ can be covered by coordinate patches each pair of which overlap positively (if they overlap at all), $M$ is **orientable**; otherwise **non-orientable**. Given such a covering, adjoin all other patches overlapping these positively; the expanded collection still overlaps positively and is an **orientation** on $M$. A manifold with an orientation is an **oriented manifold**.

This makes no sense for 0-manifolds; see §37. For a $k$-dimensional vector space $V$ we now have two notions of orientation; they correspond: given an orientation of $V$ in the sense of §20, for each frame $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ in it the linear isomorphism $\alpha : \mathbb{R}^k \to V$ with $\alpha(\mathbf{e}_i) = \mathbf{v}_i$ is a coordinate patch on $V$; any two such overlap positively, and they specify an orientation in the present sense.

## 34.1 Oriented manifolds in $\mathbb{R}^n$ of dimensions 1, $n-1$ and $n$

**Definition 34.3 (unit tangent field)**

Let $M$ be an oriented 1-manifold in $\mathbb{R}^n$. For $\mathbf{p} \in M$ choose a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation, with $\alpha(t_0) = \mathbf{p}$, and define

$$T(\mathbf{p}) = (\mathbf{p}; D\alpha(t_0)/\|D\alpha(t_0)\|).$$

$T$ is the **unit tangent field corresponding to the orientation** of $M$.

$(\mathbf{p}; D\alpha(t_0))$ is the velocity of the curve $\alpha$ at $t_0$, and $T(\mathbf{p})$ is this vector normalized. $T$ is well-defined: if $\beta$ is another patch in the orientation with $\mathbf{p} = \beta(t_1)$ and $g = \beta^{-1} \circ \alpha$, then $D\alpha(t_0) = D\beta(t_1) \cdot Dg(t_0)$ with $Dg(t_0) > 0$ (orientation-preserving), so $D\alpha(t_0)/\|D\alpha(t_0)\| = D\beta(t_1)/\|D\beta(t_1)\|$. $T$ is $C^\infty$ since $t_0 = \alpha^{-1}(\mathbf{p})$ is a $C^\infty$ function of $\mathbf{p}$.

**Example 34.1**

An oriented 1-manifold $M$ with unit tangent field $T$ is pictured by drawing arrows along $M$ in the direction of $T$; this is a *directed curve* in calculus.

If $M$ has boundary points $\mathbf{p}, \mathbf{q}$ (e.g. an arc), a patch $\alpha : U \to V$ about the boundary point $\mathbf{p}$ has $U$ open in $\mathbb{H}^1$, so $T(\mathbf{p})$ must point *into* $M$ from $\mathbf{p}$; likewise at $\mathbf{q}$. Then no unit tangent field points into $M$ at both ends, and $M$ would seem non-orientable—an anomaly. It disappears if we also allow patches with domains open in the left half-line $\mathbb{L}^1 = \{x : x \leq 0\}$.

**Convention**

In the case of a 1-manifold $M$, we allow the domains of the coordinate patches on $M$ to be open sets in $\mathbb{R}^1$ or in $\mathbb{H}^1$ or in $\mathbb{L}^1$.

With this, every 1-manifold is orientable (we shall not prove this).

**Definition 34.4 (unit normal field)**

Let $M$ be an $(n-1)$-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Let $(\mathbf{p}; \mathbf{n})$ be a unit vector in $\mathcal{T}_\mathbf{p}(\mathbb{R}^n)$ orthogonal to the $(n-1)$-dimensional subspace $\mathcal{T}_\mathbf{p}(M)$; $\mathbf{n}$ is determined up to sign. Given an orientation of $M$, choose a patch $\alpha : U \to V$ in it with $\alpha(\mathbf{x}) = \mathbf{p}$; the columns $\partial\alpha/\partial x_i$ give a basis of $\mathcal{T}_\mathbf{p}(M)$. Fix the sign of $\mathbf{n}$ by requiring the frame $(\mathbf{n}, \partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_{n-1})$ to be right-handed, i.e. $\det[\mathbf{n}\ D\alpha(\mathbf{x})] > 0$. The field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ is the **unit normal field to $M$ corresponding to the orientation** of $M$. (It is well-defined and $C^\infty$; see §38.)

**Example 34.2 (Möbius band)**

The Möbius band in $\mathbb{R}^3$ has no continuous unit normal vector field; it is not orientable.

**Example 34.3 (Klein bottle)**

The Klein bottle $K$ can be pictured in $\mathbb{R}^3$ as a self-intersecting surface swept out by a moving circle $C_0, C_1, C_2, \ldots$. It can be represented as a 2-manifold *without* self-intersections in $\mathbb{R}^4$: keep the circle in $\mathbb{R}^3 \times 0$, but as it approaches the spot where it would cross part of the surface already generated, let it move "up" into $\mathbb{R}^3 \times \mathbb{H}^1_+$ until past that spot, then come back down. $K$ contains a copy of the Möbius band $M$; if $K$ were orientable, so would be $M$ (take all patches on $M$ that overlap positively the patches of the orientation of $K$). So $K$ is not orientable.

**Definition 34.5 (natural orientation)**

Let $M$ be an $n$-manifold in $\mathbb{R}^n$. For a patch $\alpha : U \to V$ on $M$, $D\alpha$ is $n$ by $n$. The **natural orientation** of $M$ consists of all coordinate patches $\alpha$ on $M$ with $\det D\alpha > 0$; any two overlap positively.

$M$ can be covered by such patches: given $\mathbf{p}$ and a patch $\alpha : U \to V$ about $\mathbf{p}$ with $U$ open in $\mathbb{R}^n$ or $\mathbb{H}^n$, shrink $U$ to an open $\epsilon$-ball or its intersection with $\mathbb{H}^n$; then $U$ is connected and $\det D\alpha$ has constant sign on $U$. If negative, replace $\alpha$ by $\alpha \circ r$ with $r(x_1, \ldots, x_n) = (-x_1, x_2, \ldots, x_n)$.

## 34.2 Reversing the orientation of a manifold

**Definition 34.6 (reverse orientation)**

Let $r : \mathbb{R}^k \to \mathbb{R}^k$ be the reflection $r(x_1, \dots, x_k) = (-x_1, x_2, \dots, x_k)$; it is its own inverse, carries $\mathbb{H}^k$ to itself if $k > 1$, and $\mathbb{H}^1$ to $\mathbb{L}^1$ if $k = 1$. If $M$ is an oriented $k$-manifold and $\alpha_i : U_i \to V_i$ are the patches of its orientation, let $\beta_i = \alpha_i \circ r : r(U_i) \to V_i$. Then $\beta_i$ overlaps $\alpha_i$ negatively, while the $\beta_i$ overlap each other positively; they form the **reverse** (or **opposite**) orientation.

So every orientable $k$-manifold has at least two orientations; a connected one has exactly two. (A 1-manifold with four components, for example, has four orientations.) If $M$ is an oriented 1-manifold with tangent field $T$, reversing the orientation replaces $T$ by $-T$, since $(\alpha \circ r)(t) = \alpha(-t)$. If $M$ is an oriented $(n-1)$-manifold with normal field $N$, reversing replaces $N$ by $-N$: $\partial(\alpha \circ r)/\partial x_1 = -\partial\alpha/\partial x_1$, $\partial(\alpha \circ r)/\partial x_i = \partial\alpha/\partial x_i$ for $i > 1$, and $(\mathbf{n}, \partial\alpha/\partial x_1, \dots)$ is right-handed iff $(-\mathbf{n}, -\partial\alpha/\partial x_1, \partial\alpha/\partial x_2, \dots)$ is.

## 34.3 The induced orientation of $\partial M$

**Theorem 34.1**

Let $k > 1$. If $M$ is an orientable $k$-manifold with non-empty boundary, then $\partial M$ is orientable.

**Proof.** Let $\mathbf{p} \in \partial M$ and $\alpha : U \to V$ a patch about $\mathbf{p}$. The restricted patch on $\partial M$ (§24) is $\alpha_0 = \alpha \circ b$, where $b : \mathbb{R}^{k-1} \to \mathbb{R}^k$, $b(x_1, \dots, x_{k-1}) = (x_1, \dots, x_{k-1}, 0)$. We show: if $\alpha, \beta$ are patches about $\mathbf{p}$ overlapping positively, then so do $\alpha_0, \beta_0$. Let $g = \beta^{-1} \circ \alpha : W_0 \to W_1$, with $W_0, W_1$ open in $\mathbb{H}^k$; then $\det Dg > 0$. If $\mathbf{x} \in \partial\mathbb{H}^k$, the last row of $Dg(\mathbf{x})$ is $Dg_k = [0 \ \cdots \ 0 \ \partial g_k/\partial x_k]$ with $\partial g_k/\partial x_k \geq 0$: incrementing one of $x_1, \dots, x_{k-1}$ from a point of $\partial\mathbb{H}^k$ does not change $g_k$ (which stays 0, as $g$ maps $\partial\mathbb{H}^k$ into $\partial\mathbb{H}^k$), while incrementing $x_k$ positively increases $g_k$ (as $g$ maps into $\mathbb{H}^k$). Since $\det Dg \neq 0$, $\partial g_k/\partial x_k > 0$, and since $\det Dg > 0$,

$$\det \frac{\partial(g_1, \dots, g_{k-1})}{\partial(x_1, \dots, x_{k-1})} > 0.$$

This matrix is the derivative of the transition function $\beta_0^{-1} \circ \alpha_0$. ■

So an orientation of $M$ gives one of $\partial M$ by restricting the patches of the orientation; but this is not always the one we prefer:

**Definition 34.7 (induced orientation of $\partial M$)**

Let $M$ be an orientable $k$-manifold with non-empty boundary and fix an orientation of $M$. The **induced orientation** of $\partial M$ is: if $k$ is even, the orientation obtained by restricting the patches of the orientation of $M$; if $k$ is odd, the opposite of that orientation.

**Example 34.4**

$S^2$ and the torus $T$ are orientable 2-manifolds, being boundaries of orientable 3-manifolds in $\mathbb{R}^3$. If $M$ is a 3-manifold in $\mathbb{R}^3$ oriented naturally, the induced orientation of $\partial M$ corresponds to the unit normal field pointing *outwards* from $M$ (proved in §38). Informally: let $\alpha$ be a patch about $\mathbf{p} \in \partial M$ in the natural orientation; $(\alpha \circ b)(\mathbf{x}) = \alpha(x_1, x_2, 0)$ is the restricted patch. Since $\dim M = 3$ is odd, the induced orientation of $\partial M$ is opposite to that of the restrictions, so the normal $N = (\mathbf{p}; \mathbf{n})$ for the induced orientation satisfies: $(-\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-

handed. Since $\det D\alpha > 0$, $(\partial\alpha/\partial x_3, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed. So $-\mathbf{n}$ and $\partial\alpha/\partial x_3$ lie on the same side of the tangent plane. Since $\partial\alpha/\partial x_3$ points *into* $M$, $\mathbf{n}$ points outwards.

**Example 34.5**

Let $M$ be an oriented 2-manifold in $\mathbb{R}^3$ with non-empty boundary, $N$ the unit normal field for the orientation of $M$, and $T$ the unit tangent field for the induced orientation of $\partial M$. For $\mathbf{p} \in \partial M$ let $W(\mathbf{p})$ be the unit vector perpendicular to $N(\mathbf{p})$ and $T(\mathbf{p})$ with $(N, T, W)$ right-handed. Then $W(\mathbf{p})$ is tangent to $M$ and points into $M$. (This makes precise the calculus description: walking around $\partial M$ in the direction of $T$ with head pointing along $N$, $M$ is on your left.) Indeed, let $\alpha : U \to V$ be a patch about $\mathbf{p}$ in the orientation of $M$; since $\dim M = 2$ is even, $\alpha \circ b$ belongs to the induced orientation of $\partial M$, so $\partial\alpha/\partial x_1$ points in the direction of $T$. $\partial\alpha/\partial x_2$ is the velocity of a curve moving into $M$, so it points into $M$; choose $\lambda$ with $\mathbf{w} = \partial\alpha/\partial x_2 + \lambda\,\partial\alpha/\partial x_1$ orthogonal to $\partial\alpha/\partial x_1$; then $\mathbf{w}$ points into $M$, and $W(\mathbf{p}) = (\mathbf{p}; \mathbf{w}/\|\mathbf{w}\|)$. By definition $(\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed, and $\det[\mathbf{n}\ \partial\alpha/\partial x_1\ \partial\alpha/\partial x_2] = \det[\mathbf{n}\ \partial\alpha/\partial x_1\ \mathbf{w}]$; so $(N, T, W)$ is right-handed.

## §35. Integrating Forms over Oriented Manifolds

**Motivation.** With orientations fixed, the patch-by-patch definition with partitions of unity gives a well-defined integral of a $k$-form over an oriented compact $k$-manifold.

The procedure parallels §25, so we abbreviate the details.

**Definition 35.1 (integral, support in one patch)**

Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$ and $\omega$ a $k$-form defined in an open set of $\mathbb{R}^n$ containing $M$. Let $C = M \cap (\text{Support}\,\omega)$, a compact set. Suppose there is a coordinate patch $\alpha : U \to V$ on $M$ belonging to the orientation of $M$ with $C \subset V$; we may assume $U$ bounded. Define

$$\int_M \omega = \int_{\text{Int}\,U} \alpha^*\omega,$$

where $\text{Int}\,U = U$ if $U$ is open in $\mathbb{R}^k$ and $\text{Int}\,U = U \cap \mathbb{H}^k_+$ if $U$ is open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$.

*Existence*: $\alpha$ extends to a $C^\infty$ map on an open set $U'$ of $\mathbb{R}^k$, and $\alpha^*\omega$ extends to a $C^\infty$ form $h\,dx_1 \wedge \cdots \wedge dx_k$ on $U'$; then $\int_{\text{Int}\,U} \alpha^*\omega = \int_{\text{Int}\,U} h$. $h$ is continuous on $U$ and vanishes outside the compact set $\alpha^{-1}(C)$, so it is bounded; it vanishes near each point of $\text{Bd}\,U$ (if $U$ is open in $\mathbb{R}^k$), or near each point of $\text{Bd}\,U$ not in $\partial\mathbb{H}^k$, a set of measure zero. So $h$ is integrable over $U$ and over $\text{Int}\,U$. *Well-definedness*: as in Lemma 25.1, using now that transition functions between patches of the orientation are orientation-preserving, so the sign in Theorem 33.1 is $+$. *Linearity*: if $\omega$ and $\eta$ have supports whose intersections with $M$ lie in a single patch of the orientation, $\int_M a\omega + b\eta = a\int_M \omega + b\int_M \eta$, since $\alpha^*$ and $\int_{\text{Int}\,U}$ are linear. *Orientation*: if $-M$ denotes $M$ with the opposite orientation, $\int_{-M} \omega = -\int_M \omega$, by Theorem 33.1.

**Definition 35.2 (integral of a form over a compact oriented manifold)**

Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$ and $\omega$ a $k$-form defined in an open set of $\mathbb{R}^n$ containing $M$. Cover $M$ by coordinate patches belonging to its orientation, and choose a

partition of unity $\phi_1, \dots, \phi_l$ on $M$ dominated by this collection (Lemma 25.2). Define

$$\int_M \omega = \sum_{i=1}^{l} \Big[ \int_M \phi_i \omega \Big].$$

This agrees with Definition 35.1 when $M \cap \operatorname{Support} \omega$ lies in one patch, by linearity and $\omega(\mathbf{x}) = \sum_i \phi_i(\mathbf{x})\omega(\mathbf{x})$ for $\mathbf{x} \in M$; it is independent of the partition of unity by the argument used for $\int_M f\, dV$.

**Theorem 35.1**

Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$. Let $\omega, \eta$ be $k$-forms defined in an open set of $\mathbb{R}^n$ containing $M$. Then

$$\int_M (a\omega + b\eta) = a \int_M \omega + b \int_M \eta.$$

If $-M$ denotes $M$ with the opposite orientation, then $\int_{-M} \omega = -\int_M \omega$.

**Proof [Supplement].** Apply the single-patch facts above to each $\phi_i\omega$, $\phi_i\eta$ and sum; for $-M$, the patches $\alpha_i \circ r$ of the reverse orientation and the same partition of unity may be used. ■

As with $\int_M f\, dV$, in practice one evaluates $\int_M \omega$ by breaking $M$ into pieces:

**Theorem 35.2**

Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$. Let $\omega$ be a $k$-form defined in an open set of $\mathbb{R}^n$ containing $M$. Suppose that $\alpha_i : A_i \to M_i$, for $i = 1, \dots, N$, is a coordinate patch on $M$ belonging to the orientation of $M$, such that $A_i$ is open in $\mathbb{R}^k$ and $M$ is the disjoint union of the open sets $M_1, \dots, M_N$ of $M$ and a set $K$ of measure zero in $M$. Then

$$\int_M \omega = \sum_{i=1}^{N} \Big[ \int_{A_i} \alpha_i^* \omega \Big].$$

**Proof [Supplement].** Copy the proof of Theorem 25.4: by linearity assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha : U \to V$ of the orientation, and write $\alpha^*\omega = F\, dx_1 \wedge \cdots \wedge dx_k$, $\alpha_i^*\omega = F_i\, dx_1 \wedge \cdots \wedge dx_k$. With $W_i = \alpha^{-1}(M_i \cap V)$ and $L = \alpha^{-1}(K \cap V)$, $\int_M \omega = \int_{\operatorname{Int} U} F = \sum_i \int_{W_i} F$ as in Step 1 there. The transition map $\alpha_i^{-1} \circ \alpha : W_i \to B_i = \alpha_i^{-1}(M_i \cap V)$ is an orientation-preserving diffeomorphism, so by Theorem 33.1 (sign $+$) $\int_{W_i} F = \int_{B_i} F_i$, and $\int_{B_i} F_i = \int_{A_i} F_i$ by additivity as in Step 2 there. (Alternatively combine Theorems 25.4 and 36.2.) ■

## §36. A Geometric Interpretation of Forms and Integrals

**Motivation.** Integrals of forms are related to the scalar integrals of Chapter 5 via unit normal and tangent fields. This explains how line integrals, flux integrals and volume integrals of vector calculus are all integrals of forms.

The integral of a $k$-form over an oriented $k$-manifold is abstract; we relate it to the integral of a scalar function with respect to volume. First, alternating tensors versus volume:

**Theorem 36.1**

Let $W$ be a $k$-dimensional linear subspace of $\mathbb{R}^n$; let $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ be an orthonormal $k$-frame in $W$, and let $f$ be an alternating $k$-tensor on $W$. If $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is an arbitrary $k$-tuple in $W$, then

$$f(\mathbf{x}_1, \dots, \mathbf{x}_k) = \epsilon\, V(\mathbf{x}_1, \dots, \mathbf{x}_k)\, f(\mathbf{a}_1, \dots, \mathbf{a}_k),$$

where $\epsilon = \pm 1$. If the $\mathbf{x}_i$ are independent, then $\epsilon = +1$ if the frames $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ and $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ belong to the same orientation of $W$ and $\epsilon = -1$ otherwise.

(If the $\mathbf{x}_i$ are dependent, $V(\mathbf{x}_1, \dots, \mathbf{x}_k) = 0$ by Theorem 21.3 and $\epsilon$ does not matter.)

**Proof.**

*Step 1.* $W = \mathbb{R}^k$. Then $f$ is a multiple of the determinant: $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c \det[\mathbf{x}_1 \cdots \mathbf{x}_k]$. If the $\mathbf{x}_i$ are dependent, $f$ vanishes. Otherwise $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c\,\epsilon_1 V(\mathbf{x}_1, \dots, \mathbf{x}_k)$ with $\epsilon_1 = +1$ iff $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is right-handed, and $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2 V(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2$ similarly. So $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = (\epsilon_1/\epsilon_2) V(\mathbf{x}_1, \dots, \mathbf{x}_k) f(\mathbf{a}_1, \dots, \mathbf{a}_k)$, and $\epsilon_1/\epsilon_2 = +1$ iff the two frames belong to the same orientation.

*Step 2.* In general choose an orthogonal transformation $h$ of $\mathbb{R}^n$ carrying $W$ onto $\mathbb{R}^k \times 0$, with inverse $k : \mathbb{R}^k \times 0 \to W$. Then $k^* f$ is an alternating tensor on $\mathbb{R}^k \times 0$; $(h(\mathbf{x}_i))$ is a $k$-tuple and $(h(\mathbf{a}_i))$ an orthonormal frame in $\mathbb{R}^k \times 0$. By Step 1, $(k^* f)(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k)) = \epsilon V(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k))(k^* f)(h(\mathbf{a}_1), \dots, h(\mathbf{a}_k))$. Since $V$ is unchanged by orthogonal transformations, this is the desired equation. If the $\mathbf{x}_i$ are independent, $\epsilon = +1$ iff $(h(\mathbf{x}_i))$, $(h(\mathbf{a}_i))$ belong to the same orientation of $\mathbb{R}^k \times 0$, iff $(\mathbf{x}_i)$, $(\mathbf{a}_i)$ belong to the same orientation of $W$. ■

So for two orthonormal frames $(\mathbf{a}_i)$, $(\mathbf{b}_i)$ in $W$, $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = \pm f(\mathbf{b}_1, \dots, \mathbf{b}_k)$, the sign depending on whether they lie in the same orientation of $W$.

**Definition 36.1 (natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$)**

Let $M$ be an oriented $k$-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Choose a patch $\alpha : U \to V$ in the orientation of $M$ about $\mathbf{p}$, with $\alpha(\mathbf{x}) = \mathbf{p}$. The collection of $k$-frames in $\mathcal{T}_{\mathbf{p}}(M)$ of the form $(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_k))$, with $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ right-handed in $\mathbb{R}^k$, is the **natural orientation** of $\mathcal{T}_{\mathbf{p}}(M)$ induced by the orientation of $M$; it is independent of $\alpha$.

**Theorem 36.2**

Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$; let $\omega$ be a $k$-form defined in an open set of $\mathbb{R}^n$ containing $M$. Let $\lambda$ be the scalar function on $M$ defined by the equation

$$\lambda(\mathbf{p}) = \omega(\mathbf{p})((\mathbf{p}; \mathbf{a}_1), \dots, (\mathbf{p}; \mathbf{a}_k)),$$

where $((\mathbf{p}; \mathbf{a}_1), \dots, (\mathbf{p}; \mathbf{a}_k))$ is any orthonormal frame in the linear space $\mathcal{T}_{\mathbf{p}}(M)$ belonging to its natural orientation. Then $\lambda$ is continuous, and

$$\int_M \omega = \int_M \lambda\, dV.$$

**Proof.** By linearity assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha : U \to V$ of the orientation. Write $\alpha^* \omega = h\, dx_1 \wedge \cdots \wedge dx_k$ and let $\alpha(\mathbf{x}) = \mathbf{p}$. Then

$$h(\mathbf{x}) = (\alpha^* \omega)(\mathbf{x})((\mathbf{x}; \mathbf{e}_1), \dots, (\mathbf{x}; \mathbf{e}_k)) = \omega(\mathbf{p})((\mathbf{p}; \partial\alpha/\partial x_1), \dots, (\mathbf{p}; \partial\alpha/\partial x_k)) = \pm V(D\alpha(\mathbf{x}))\lambda(\mathbf{p})$$

by Theorem 36.1; the sign is + because $((\mathbf{p}; \partial\alpha/\partial x_1), \dots, (\mathbf{p}; \partial\alpha/\partial x_k))$ belongs to the natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$ by definition. Since $V(D\alpha) \neq 0$ ($D\alpha$ has rank $k$) and $\mathbf{x} = \alpha^{-1}(\mathbf{p})$ is continuous

in $\mathbf{p}$, $\lambda(\mathbf{p}) = h(\mathbf{x})/V(D\alpha(\mathbf{x}))$ is continuous. Hence

$$\int_M \lambda\, dV = \int_{\operatorname{Int} U} (\lambda \circ \alpha) V(D\alpha) = \int_{\operatorname{Int} U} h = \int_{\operatorname{Int} U} \alpha^*\omega = \int_M \omega.$$

■

So for each $k$-form $\omega$ there is a scalar function $\lambda$ (in fact $C^\infty$) with $\int_M \omega = \int_M \lambda\, dV$. The converse is true but harder: there is a $k$-form $\omega_V$, defined on an open set about $M$, whose value on any orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation is 1. Then for any $C^\infty$ function $\lambda$ on $M$, $\int_M \lambda\, dV = \int_M \lambda\omega_V$, so the integral of $\lambda$ over $M$ is the integral of the form $\lambda\omega_V$. $\omega_V$ is a **volume form** for $M$, since $\int_M \omega_V = \int_M dV = \mathrm{v}(M)$. This applies only to orientable $M$: on a non-orientable $M$ the integral of a scalar function is defined but that of a form is not.

**Notation**

Some mathematicians denote the volume form $\omega_V$ by $dV$ (or $dV$). This makes the above equations tautologies, but it can confuse: $V$ is *not* a form, and $d$ does *not* denote the differential operator in this context.

## §37. The Generalized Stokes' Theorem

This is the culmination: a theorem about integrals of forms containing the three basic theorems of vector integral calculus—Green's theorem, Stokes' theorem and the divergence theorem. Let $I^k = [0,1]^k$ be the unit $k$-cube; $\operatorname{Int} I^k = (0,1)^k$ and $\operatorname{Bd} I^k = I^k - \operatorname{Int} I^k$.

**Lemma 37.1**

Let $k > 1$. Let $\eta$ be a $k-1$ form defined in an open set $U$ of $\mathbb{R}^k$ containing the unit $k$-cube $I^k$. Assume $\eta$ vanishes at all points of $\operatorname{Bd} I^k$ except possibly at points of the subset $(\operatorname{Int} I^{k-1}) \times 0$. Then

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta,$$

where $b : I^{k-1} \to I^k$ is the map $b(u_1, \ldots, u_{k-1}) = (u_1, \ldots, u_{k-1}, 0)$.

**Proof.** Let $\mathbf{x}$ and $\mathbf{u}$ be the general points of $\mathbb{R}^k$ and $\mathbb{R}^{k-1}$. For $1 \le j \le k$ let $I_j = (1, \ldots, \hat{j}, \ldots, k)$; the elementary $(k-1)$-forms in $\mathbb{R}^k$ are $dx_{I_j} = dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_k$. By linearity of the integrals, $d$ and $b^*$, it suffices to take $\eta = f\, dx_{I_j}$.

*Step 1.* We have

$$d\eta = df \wedge dx_{I_j} = \Big(\sum_{i=1}^k D_i f\, dx_i\Big) \wedge dx_{I_j} = (-1)^{j-1}(D_j f)\, dx_1 \wedge \cdots \wedge dx_k.$$

So, using the Fubini theorem with $\mathbf{v} = (x_1, \ldots, \hat{x}_j, \ldots, x_k)$,

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^{j-1} \int_{I^k} D_j f = (-1)^{j-1} \int_{\mathbf{v} \in I^{k-1}} \int_{x_j \in I} D_j f(x_1, \ldots, x_k).$$

By the fundamental theorem of calculus the inner integral is $f(x_1, \ldots, 1, \ldots, x_k) - f(x_1, \ldots, 0, \ldots, x_k)$, with 1 and 0 in the $j$th place. $\eta$, hence $f$, vanishes on $\operatorname{Bd} I^k$ except possibly on the open bottom face $(\operatorname{Int} I^{k-1}) \times 0$. If $j < k$ both terms vanish; if $j = k$ the result is $-f(x_1, \ldots, x_{k-1}, 0)$. Hence

$$\int_{\operatorname{Int} I^k} d\eta = \begin{cases} 0 & j < k, \\ (-1)^k \int_{I^{k-1}} (f \circ b) & j = k. \end{cases}$$

*Step 2.* $Db = \begin{bmatrix} I_{k-1} \\ 0 \end{bmatrix}$, so by Theorem 32.2,

$$b^*(dx_{I_j}) = [\det Db(1,\dots,\hat{j},\dots,k)]\, du_1 \wedge \cdots \wedge du_{k-1} = \begin{cases} 0 & j < k, \\ du_1 \wedge \cdots \wedge du_{k-1} & j = k. \end{cases}$$

Hence $\int_{\operatorname{Int} I^{k-1}} b^*\eta$ is 0 if $j < k$ and $\int_{\operatorname{Int} I^{k-1}} (f \circ b)$ if $j = k$. Compare with Step 1. ■

**Theorem 37.2 (Stokes' theorem)**

Let $k > 1$. Let $M$ be a compact oriented $k$-manifold in $\mathbb{R}^n$; give $\partial M$ the induced orientation if $\partial M$ is not empty. Let $\omega$ be a $k-1$ form defined in an open set of $\mathbb{R}^n$ containing $M$. Then

$$\int_M d\omega = \int_{\partial M} \omega$$

if $\partial M$ is not empty; and $\int_M d\omega = 0$ if $\partial M$ is empty.

**Proof.**

*Step 1.* A special covering of $M$ by coordinate patches. If $\mathbf{p} \in M - \partial M$, choose a patch $\alpha : U \to V$ in the orientation of $M$ with $U$ open in $\mathbb{R}^k$, $U \supset I^k$, and $\alpha$ carrying a point of $\operatorname{Int} I^k$ to $\mathbf{p}$ (precede an arbitrary patch by a translation and a stretching of $\mathbb{R}^k$). Let $W = \operatorname{Int} I^k$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a coordinate patch in the orientation about $\mathbf{p}$, with $W$ open in $\mathbb{R}^k$. If $\mathbf{p} \in \partial M$, choose $\alpha : U \to V$ in the orientation with $U$ open in $\mathbb{H}^k$, $U \supset I^k$, carrying a point of $(\operatorname{Int} I^{k-1}) \times 0$ to $\mathbf{p}$. Let $W = (\operatorname{Int} I^k) \cup ((\operatorname{Int} I^{k-1}) \times 0)$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a patch in the orientation about $\mathbf{p}$ with $W$ open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$. In each case $\alpha$ can be extended to a $C^\infty$ map on an open set of $\mathbb{R}^k$ containing $I^k$.

*Step 2.* Since $d$ and the integrals are linear, and $\omega = \sum_i \phi_i \omega$ for a partition of unity $\{\phi_i\}$ on $M$ dominated by these patches, it suffices to prove the theorem when $C = M \cap \operatorname{Support} \omega$ is covered by a single patch $\alpha : W \to Y$ of Step 1. Then $M \cap \operatorname{Support} d\omega \subset C$ is covered too. Let $\eta = \alpha^*\omega$, extended (without change of notation) to a $C^\infty$ form on an open set of $\mathbb{R}^k$ containing $I^k$. $\eta$ vanishes at all points of $\operatorname{Bd} I^k$ except possibly points of $(\operatorname{Int} I^{k-1}) \times 0$, so Lemma 37.1 applies.

*Step 3.* Case $W = \operatorname{Int} I^k$; $Y$ is disjoint from $\partial M$. Since $\alpha^* d\omega = d\alpha^*\omega = d\eta$,

$$\int_M d\omega = \int_{\operatorname{Int} I^k} \alpha^* d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta.$$

Here $\eta$ vanishes outside $\operatorname{Int} I^k$, in particular on $I^{k-1} \times 0$, so $b^*\eta = 0$ and $\int_M d\omega = 0$. If $\partial M$ is empty, this is what we wanted. If not, $\int_{\partial M} \omega = 0$ too, since $\operatorname{Support} \omega$ is disjoint from $\partial M$.

*Step 4.* Case $W$ open in $\mathbb{H}^k$, $Y \cap \partial M \neq \emptyset$. As before, $\int_M d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta$. Now $\partial M \cap \operatorname{Support} \omega$ is covered by the patch $\beta = \alpha \circ b : \operatorname{Int} I^{k-1} \to Y \cap \partial M$ on $\partial M$, obtained by restricting $\alpha$. $\beta$ belongs to the induced orientation of $\partial M$ if $k$ is even, and to the opposite orientation if $k$ is odd; so computing $\int_{\partial M} \omega$ with $\beta$ we must reverse the sign when $k$ is odd:

$$\int_{\partial M} \omega = (-1)^k \int_{\operatorname{Int} I^{k-1}} \beta^*\omega.$$

Since $\beta^*\omega = b^*(\alpha^*\omega) = b^*\eta$, the theorem follows. ■

What if $k = 1$? If $\partial M$ is empty, one proves readily that $\int_M d\omega = 0$. If $\partial M$ is non-empty we must say what an "orientation" of a 0-manifold is and how to "integrate" a 0-form over it. First a special case.

**Definition 37.1 (arc)**

Let $M$ be a 1-manifold in $\mathbb{R}^n$. If there is a one-to-one $C^\infty$ map $\alpha : [a,b] \to M$ carrying $[a,b]$ onto $M$ with $D\alpha(t) \neq 0$ for all $t$, $M$ is a (smooth) **arc** in $\mathbb{R}^n$. If $M$ is oriented so that $\alpha|(a,b)$ belongs to the orientation, $\mathbf{p} = \alpha(a)$ is the **initial point** and $\mathbf{q} = \alpha(b)$ the **final point** of $M$.

**Theorem 37.3**

Let $M$ be a 1-manifold in $\mathbb{R}^n$; let $f$ be a 0-form defined in an open set about $M$. If $M$ is an arc with initial point $\mathbf{p}$ and final point $\mathbf{q}$, then

$$\int_M df = f(\mathbf{q}) - f(\mathbf{p}).$$

**Proof.** $\alpha : (a,b) \to M - \mathbf{p} - \mathbf{q}$ is a patch covering all of $M$ except a set of measure zero, so by Theorem 35.2, $\int_M df = \int_{(a,b)} \alpha^*(df)$. With $t$ the general point of $\mathbb{R}$, $\alpha^*(df) = d(f \circ \alpha) = D(f \circ \alpha)\, dt$, so $\int_{(a,b)} \alpha^*(df) = \int_{(a,b)} D(f \circ \alpha) = f(\alpha(b)) - f(\alpha(a))$ by the fundamental theorem of calculus. ■

**Definition 37.2 (orientation and integral for 0-manifolds)**

A compact 0-manifold $N$ in $\mathbb{R}^n$ is a finite set $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$. An **orientation** of $N$ is a function $\epsilon : N \to \{-1, 1\}$. If $f$ is a 0-form defined on an open set containing $N$, the **integral of $f$ over the oriented manifold $N$** is

$$\int_N f = \sum_{i=1}^{m} \epsilon(\mathbf{x}_i) f(\mathbf{x}_i).$$

If $M$ is an oriented 1-manifold with non-empty boundary, the **induced orientation** of $\partial M$ is defined by $\epsilon(\mathbf{p}) = -1$ if there is a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation of $M$ with $U$ open in $\mathbb{H}^1$, and $\epsilon(\mathbf{p}) = +1$ otherwise.

**Theorem 37.4 (Stokes' theorem in dimension 1)**

Let $M$ be a compact oriented 1-manifold in $\mathbb{R}^n$; give $\partial M$ the induced orientation if $\partial M$ is not empty. Let $f$ be a 0-form defined in an open set of $\mathbb{R}^n$ containing $M$. Then

$$\int_M df = \int_{\partial M} f$$

if $\partial M$ is not empty; and $\int_M df = 0$ if $\partial M$ is empty.

**Proof [Supplement].** Cover $M$ by patches $\alpha : W \to Y$ of its orientation, where $W$ is one of the intervals $(0,1)$, $[0,1)$, $(-1,0]$, and use a partition of unity dominated by them; by linearity it suffices to treat $f$ with $M \cap \operatorname{Support} f$ inside one such $Y$. Let $g = f \circ \alpha$, extended to a $C^\infty$ function near $[-1,1]$; then $\int_M df = \int_{\operatorname{Int} W} \alpha^* df = \int_{\operatorname{Int} W} g'$. If $W = (0,1)$, $g$ vanishes near 0 and 1, so $\int_0^1 g' = 0$, and $\operatorname{Support} f$ misses $\partial M$. If $W = [0,1)$, $\alpha(0) = \mathbf{p} \in \partial M$ has $\epsilon(\mathbf{p}) = -1$ and $g(1) = 0$: $\int_0^1 g' = -g(0) = -f(\mathbf{p}) = \int_{\partial M} f$. If $W = (-1,0]$, $\alpha(0) = \mathbf{p}$ has $\epsilon(\mathbf{p}) = +1$ (no patch of the orientation about $\mathbf{p}$ has domain open in $\mathbb{H}^1$, since such a patch would overlap $\alpha$ negatively), and $\int_{-1}^0 g' = g(0) = f(\mathbf{p}) = \int_{\partial M} f$. ■

# §38. Applications to Vector Analysis

By §31, forms of order $k = 0, 1, n-1, n$ in $\mathbb{R}^n$ can be interpreted as scalar or vector fields. We show integrals of forms can be similarly interpreted, so that Stokes' theorem yields the classical theorems of vector integral calculus.

## 38.1 The gradient theorem for 1-manifolds in $\mathbb{R}^n$

**Lemma 38.1**

Let $M$ be a compact oriented 1-manifold in $\mathbb{R}^n$; let $T$ be the unit tangent vector to $M$ corresponding to the orientation. Let $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x})) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ be a vector field defined in an open set of $\mathbb{R}^n$ containing $M$; it corresponds to the 1-form $\omega = \sum f_i\, dx_i$. Then

$$\int_M \omega = \int_M \langle F, T\rangle\, ds.$$

Here the classical $ds$ (arc length) replaces $dV$. Replacing $M$ by $-M$ changes the sign of $\int_M \omega$ and replaces $T$ by $-T$, so the right side changes sign too.

**Proof.** *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda\, ds$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation; here $T(\mathbf{p}) = (\mathbf{p}; \mathbf{t})$ is such a basis, and $\omega(\mathbf{p})(\mathbf{p}; \mathbf{t}) = \sum f_i(\mathbf{p})t_i$. So $\lambda(\mathbf{p}) = \langle F(\mathbf{p}), T(\mathbf{p})\rangle$.

*Second proof.* By linearity assume $C = M \cap \operatorname{Support}\omega$ lies in a single patch $\alpha : U \to V$ of the orientation. With $t$ the general point of $\mathbb{R}$, $\alpha^*\omega = \sum_i (f_i \circ \alpha)\, d\alpha_i = \sum_i (f_i \circ \alpha)(D\alpha_i)\, dt = \langle f \circ \alpha, D\alpha\rangle\, dt$, so $\int_M \omega = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle$. On the other hand

$$\int_M \langle F, T\rangle\, ds = \int_{\operatorname{Int} U} \langle F \circ \alpha, T \circ \alpha\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha/\|D\alpha\|\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle,$$

since $V(D\alpha) = [\det(D\alpha^{\mathrm{tr}} D\alpha)]^{1/2} = \|D\alpha\|$. ■

**Theorem 38.2 (The gradient theorem)**

Let $M$ be a compact 1-manifold in $\mathbb{R}^n$; let $T$ be a unit tangent vector field to $M$. Let $f$ be a $C^\infty$ function defined in an open set about $M$. If $\partial M$ is empty, then $\int_M \langle \operatorname{grad} f, T\rangle\, ds = 0$. If $\partial M$ consists of the points $\mathbf{x}_1, \ldots, \mathbf{x}_m$, let $\epsilon_i = -1$ if $T$ points into $M$ at $\mathbf{x}_i$ and $\epsilon_i = +1$ otherwise. Then

$$\int_M \langle \operatorname{grad} f, T\rangle\, ds = \sum_{i=1}^m \epsilon_i f(\mathbf{x}_i).$$

**Proof.** The 1-form $df$ corresponds to $\operatorname{grad} f$ (Theorem 31.1), so $\int_M df = \int_M \langle \operatorname{grad} f, T\rangle\, ds$ by the lemma. Apply Stokes' theorem in dimension 1. ■

## 38.2 The divergence theorem for $n-1$ manifolds in $\mathbb{R}^n$

Recall from §34 the unit normal field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ of an oriented $(n-1)$-manifold $M$, defined by: $\mathbf{n}$ is a unit vector orthogonal to $\mathcal{T}_{\mathbf{p}}(M)$ and $(\mathbf{n}, \partial\alpha/\partial x_1, \ldots, \partial\alpha/\partial x_{n-1})$ is right-handed for a patch $\alpha$ in the orientation with $\alpha(\mathbf{x}) = \mathbf{p}$. *$N$ is well-defined*: if $\beta$ is another such patch, $g = \beta^{-1} \circ \alpha$ and $g(\mathbf{x}) = \mathbf{y}$, then $D\alpha(\mathbf{x}) = D\beta(\mathbf{y})Dg(\mathbf{x})$, so for any $\mathbf{v} \in \mathbb{R}^n$,

$$[\mathbf{v}\ D\alpha(\mathbf{x})] = [\mathbf{v}\ D\beta(\mathbf{y})] \cdot \begin{bmatrix} 1 & 0 \\ 0 & Dg(\mathbf{x}) \end{bmatrix}, \qquad \det[\mathbf{v}\ D\alpha(\mathbf{x})] = \det[\mathbf{v}\ D\beta(\mathbf{y})] \cdot \det Dg(\mathbf{x}).$$

Since $\det Dg > 0$, $[\mathbf{v}\ D\alpha(\mathbf{x})]$ has positive determinant iff $[\mathbf{v}\ D\beta(\mathbf{y})]$ does. To see $N$ is $C^\infty$ we find a formula for it.

**Example 38.1**

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, calculus teaches that $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ is perpendicular to both, $(\mathbf{c}, \mathbf{a}, \mathbf{b})$ is right-handed, and $\|\mathbf{c}\| = V(\mathbf{a}, \mathbf{b})$; here $c_1 = \det \begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$, $c_2 = -\det \begin{bmatrix} a_1 & b_1 \\ a_3 & b_3 \end{bmatrix}$, $c_3 = \det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$. So for an oriented 2-manifold in $\mathbb{R}^3$ and a patch $\alpha$ in its orientation, $\mathbf{n} = \mathbf{c}/\|\mathbf{c}\|$ with $\mathbf{c} = \partial\alpha/\partial x_1 \times \partial\alpha/\partial x_2$ is the unit normal.

**Lemma 38.3**

Given independent vectors $\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}$ in $\mathbb{R}^n$, let $X$ be the $n$ by $n-1$ matrix $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}]$, and let $\mathbf{c}$ be the vector $\mathbf{c} = \sum c_i \mathbf{e}_i$, where

$$c_i = (-1)^{i-1} \det X(1, \ldots, \hat{\imath}, \ldots, n).$$

The vector $\mathbf{c}$ has the following properties:
(1) $\mathbf{c}$ is non-zero and orthogonal to each $\mathbf{x}_i$.
(2) The frame $(\mathbf{c}, \mathbf{x}_1, \ldots, \mathbf{x}_{n-1})$ is right-handed.
(3) $\|\mathbf{c}\| = V(X)$.

**Proof.** For $\mathbf{a} \in \mathbb{R}^n$, expanding by cofactors of the first column,

$$\det[\mathbf{a}\ \mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}] = \sum_{i=1}^{n} a_i (-1)^{i-1} \det X(1, \ldots, \hat{\imath}, \ldots, n) = \langle \mathbf{a}, \mathbf{c} \rangle.$$

(1) With $\mathbf{a} = \mathbf{x}_i$ the matrix has two equal columns, so $\langle \mathbf{x}_i, \mathbf{c} \rangle = 0$. Since the columns of $X$ span an $(n-1)$-dimensional space, so do its rows; some $n-1$ rows are independent, say all but the $i$th; then $c_i \neq 0$.

(2) With $\mathbf{a} = \mathbf{c}$, $\det[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \|\mathbf{c}\|^2 > 0$.

(3) This follows from Theorem 21.4; alternatively,

$$[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}]^{\mathrm{tr}} \cdot [\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \begin{bmatrix} \|\mathbf{c}\|^2 & 0 \\ 0 & X^{\mathrm{tr}} X \end{bmatrix},$$

and taking determinants with (2), $\|\mathbf{c}\|^4 = \|\mathbf{c}\|^2 V(X)^2$. ■

**Corollary 38.4**

If $M$ is an oriented $n-1$ manifold in $\mathbb{R}^n$, then the unit normal vector $N(\mathbf{p})$ corresponding to the orientation of $M$ is a $C^\infty$ function of $\mathbf{p}$.

**Proof.** For a patch $\alpha : U \to V$ in the orientation let $c_i(\mathbf{x}) = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)(\mathbf{x})$ and $\mathbf{c}(\mathbf{x}) = \sum c_i(\mathbf{x}) \mathbf{e}_i$. Then $N(\mathbf{p}) = (\mathbf{p}; \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ with $\mathbf{x} = \alpha^{-1}(\mathbf{p})$, a $C^\infty$ function of $\mathbf{p}$. ■

**Lemma 38.5**

Let $M$ be a compact oriented $n-1$ manifold in $\mathbb{R}^n$; let $N$ be the corresponding unit normal vector field. Let $G$ be a vector field defined in an open set $U$ of $\mathbb{R}^n$ containing $M$. If we denote the general point of $\mathbb{R}^n$ by $\mathbf{y}$, this vector field has the form $G(\mathbf{y}) = (\mathbf{y}; g(\mathbf{y})) = (\mathbf{y}; \sum g_i(\mathbf{y}) \mathbf{e}_i)$;

it corresponds to the $n-1$ form

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} g_i \, dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n.$$

Then $\int_M \omega = \int_M \langle G, N \rangle \, dV$.

Replacing $M$ by $-M$ changes the sign of $\int_M \omega$ and replaces $N$ by $-N$.

**Proof.** *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda \, dV$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis $((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1}))$ of $\mathcal{T}_\mathbf{p}(M)$ in its natural orientation. Let $A = [\mathbf{a}_1 \cdots \mathbf{a}_{n-1}]$ and $\mathbf{c} = \sum c_i \mathbf{e}_i$ with $c_i = (-1)^{i-1} \det A(1, \ldots, \hat{\imath}, \ldots, n)$. By Lemma 38.3, $\mathbf{c}$ is orthogonal to each $\mathbf{a}_i$, $(\mathbf{c}, \mathbf{a}_1, \ldots, \mathbf{a}_{n-1})$ is right-handed, and $\|\mathbf{c}\| = V(A) = [\det(A^{\mathrm{tr}} A)]^{1/2} = [\det I_{n-1}]^{1/2} = 1$. So $N = (\mathbf{p}; \mathbf{c})$. By Theorem 27.7, $dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1})) = \det A(1, \ldots, \hat{\imath}, \ldots, n)$, hence $\lambda(\mathbf{p}) = \sum_i (-1)^{i-1} g_i(\mathbf{p}) \det A(1, \ldots, \hat{\imath}, \ldots, n) = \sum_i g_i(\mathbf{p}) c_i = \langle G, N \rangle$.

*Second proof.* By linearity assume $C = M \cap \operatorname{Support} \omega$ lies in a single patch $\alpha : U \to V$ of the orientation. By Theorem 32.2,

$$\int_M \omega = \int_{\operatorname{Int} U} \alpha^* \omega = \int_{\operatorname{Int} U} \Big[ \sum_{i=1}^{n} (-1)^{i-1} (g_i \circ \alpha) \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big].$$

With $c_i = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)$, $N(\alpha(\mathbf{x})) = (\alpha(\mathbf{x}); \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ as in Corollary 38.4, and

$$\int_M \langle G, N \rangle \, dV = \int_{\operatorname{Int} U} \langle G \circ \alpha, N \circ \alpha \rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle g \circ \alpha, \mathbf{c} \rangle = \int_{\operatorname{Int} U} \Big[ \sum_i (g_i \circ \alpha)(-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big],$$

since $\|\mathbf{c}\| = V(D\alpha)$. ■

**Lemma 38.6**

Let $M$ be a compact $n$-manifold in $\mathbb{R}^n$, oriented naturally. Let $\omega = h \, dx_1 \wedge \cdots \wedge dx_n$ be an $n$-form defined in an open set of $\mathbb{R}^n$ containing $M$. Then $h$ is the corresponding scalar field, and

$$\int_M \omega = \int_M h \, dV.$$

**Proof.** *First proof.* $\int_M \omega = \int_M \lambda \, dV$ with $\lambda$ the value of $\omega$ on an orthonormal basis of $\mathcal{T}_\mathbf{p}(M) = \mathcal{T}_\mathbf{p}(\mathbb{R}^n)$ in its natural orientation. A patch $\alpha$ of the natural orientation has $\det D\alpha > 0$, so the natural orientation of $\mathcal{T}_\mathbf{p}(M)$ consists of the right-handed frames; the usual basis is one, and the value of $\omega$ on it is $h$. *Second proof.* Assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha$ of the orientation. By definition $\int_M \omega = \int_{\operatorname{Int} U} \alpha^* \omega = \int_{\operatorname{Int} U} (h \circ \alpha) \det D\alpha$ and $\int_M h \, dV = \int_{\operatorname{Int} U} (h \circ \alpha) V(D\alpha)$; and $V(D\alpha) = |\det D\alpha| = \det D\alpha$ since $\alpha$ belongs to the natural orientation. ■

$\int_M h \, dV$ equals the ordinary integral of $h$ over the bounded set $M$: $A = M - \partial M$ is open in $\mathbb{R}^n$ and the identity $i : A \to A$ is a patch of the natural orientation covering $M$ except a set of measure zero, so by Theorem 25.4, $\int_M h \, dV = \int_A (h \circ i) V(Di) = \int_A h$, which equals $\int_M h$ since $\partial M$ has measure zero in $\mathbb{R}^n$.

For a naturally oriented $n$-manifold $M$ in $\mathbb{R}^n$, the induced orientation of $\partial M$ behaves as in Example 34.4 for $n = 3$. The **inward normal** to $\partial M$ at $\mathbf{p}$ is the velocity vector of a curve that begins at $\mathbf{p}$ and moves into $M$ as the parameter increases; the **outward normal** is its negative.

**Lemma 38.7**

Let $M$ be an $n$-manifold in $\mathbb{R}^n$. If $M$ is oriented naturally, then the induced orientation of $\partial M$ corresponds to the unit normal field $N$ to $\partial M$ that points outwards from $M$ at each point of $\partial M$.

**Proof.** Let $\alpha : U \to V$ be a patch about $\mathbf{p} \in \partial M$ in the orientation of $M$, so $\det D\alpha > 0$, and let $b(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0)$. Then $\alpha_0 = \alpha \circ b$ is a patch on $\partial M$ about $\mathbf{p}$; it belongs to the induced orientation of $\partial M$ if $n$ is even and to its opposite if $n$ is odd. Let $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n})$ be the normal for the induced orientation. Then $\det[(-1)^n \mathbf{n} \;\; D\alpha_0] > 0$, which implies

$$\det[D\alpha_0 \;\; \mathbf{n}] = \det\left[\frac{\partial \alpha}{\partial x_1} \;\; \cdots \;\; \frac{\partial \alpha}{\partial x_{n-1}} \;\; \mathbf{n}\right] < 0,$$

(moving the first column to the last position changes the sign by $(-1)^{n-1}$). On the other hand $\det D\alpha = \det[\frac{\partial \alpha}{\partial x_1} \cdots \frac{\partial \alpha}{\partial x_{n-1}} \;\; \frac{\partial \alpha}{\partial x_n}] > 0$. So $\mathbf{n}$ and $\partial\alpha/\partial x_n$ lie on opposite sides of the tangent plane $\mathcal{T}_{\mathbf{p}}(\partial M)$. $\partial\alpha/\partial x_n$ is the velocity vector of a curve beginning at a point of $\partial M$ and moving into $M$ as the parameter increases; thus $\mathbf{n}$ is the outward normal. ■

**Theorem 38.8 (The divergence theorem)**

Let $M$ be a compact $n$-manifold in $\mathbb{R}^n$. Let $N$ be the unit normal vector field to $\partial M$ that points outwards from $M$. If $G$ is a vector field defined in an open set of $\mathbb{R}^n$ containing $M$, then

$$\int_M (\operatorname{div} G)\, dV = \int_{\partial M} \langle G, N \rangle \, dV.$$

The left side is an integral with respect to $n$-volume, the right with respect to $(n-1)$-volume.

**Proof.** Let $\omega = \beta_{n-1} G$ be the corresponding $(n-1)$-form. Orient $M$ naturally and give $\partial M$ the induced orientation; by Lemma 38.7 the normal field corresponding to this orientation is $N$, so $\int_{\partial M} \omega = \int_{\partial M} \langle G, N \rangle \, dV$ by Lemma 38.5. By Theorem 31.1, $d\omega = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$, and Lemma 38.6 gives $\int_M d\omega = \int_M (\operatorname{div} G)\, dV$. Apply Stokes' theorem. ■

In $\mathbb{R}^3$ the divergence theorem is sometimes called *Gauss' theorem*.

## 38.3 Stokes' theorem for 2-manifolds in $\mathbb{R}^3$

**Theorem 38.9 (Stokes' theorem—classical version)**

Let $M$ be a compact orientable 2-manifold in $\mathbb{R}^3$. Let $N$ be a unit normal field to $M$. Let $F$ be a $C^\infty$ vector field defined in an open set about $M$. If $\partial M$ is empty, then $\int_M \langle \operatorname{curl} F, N \rangle \, dV = 0$. If $\partial M$ is non-empty, let $T$ be the unit tangent vector field to $\partial M$ chosen so that the vector $W(\mathbf{p}) = N(\mathbf{p}) \times T(\mathbf{p})$ points into $M$ from $\partial M$. Then

$$\int_M \langle \operatorname{curl} F, N \rangle \, dV = \int_{\partial M} \langle F, T \rangle \, ds.$$

**Proof.** Let $\omega = \alpha_1 F$ be the corresponding 1-form; by Theorem 31.2, $\operatorname{curl} F$ corresponds to the 2-form $d\omega$. Orient $M$ so that $N$ is its unit normal field; by Lemma 38.5, $\int_M d\omega = \int_M \langle \operatorname{curl} F, N \rangle \, dV$. If $\partial M$ is non-empty, its induced orientation corresponds to the unit tangent field $T$ (Example 34.5: $(N, T, W)$ is right-handed, i.e. $W = N \times T$ points into $M$), so Lemma 38.1 gives $\int_{\partial M} \omega = \int_{\partial M} \langle F, T \rangle \, ds$. Apply Stokes' theorem. ■